ECON 282E: Foundations of Macroeconomics

Heterogeneous Agents, Computation, and Artificial Intelligence

Ph.D. Elective · University of California, Riverside · Fall Quarter 2026

Meetings	Thursdays, 9:00–11:50 a.m., Sproul Hall 4128. Each meeting has two 75-minute blocks separated by a 20-minute break.
Instructor	Zhigang Feng
Email	[UCR email]
Office hours	[day, time, room], and by appointment
Course website	`https://vonzhg.github.io/Macro-2026Fall/` — slides, labs, homework, announcements, and the discussion forum

1. Course Description

Modern macroeconomics asks questions about *distributions* and *policy*: who bears income risk, how inequality interacts with the business cycle, how aging and public debt burden different cohorts, what a government should tax when it cannot commit, and how artificial intelligence will reshape work and wages. Answering these questions requires models with heterogeneous agents, frictions, and strategic governments — and those models can almost never be solved with pencil and paper.

This course builds the foundations in three steps. **On paper**, we start from the neoclassical growth model and extend it to the workhorse models of modern research: the Bewley–Huggett–Aiyagari incomplete-markets model, the Krusell–Smith model with aggregate risk, overlapping-generations models, and optimal fiscal policy with and without commitment. For each model we ask how it is built, what equilibrium means, what can be characterized, and which questions it can answer. **On the machine**, we learn to solve these models: numerical dynamic programming, programming in Python and PyTorch, function approximation, optimization, root-finding, quadrature, discretization of stochastic processes, perturbation, and projection methods — culminating in the computation of heterogeneous-agent economies and the "computational wall" they run into. **At the frontier**, we study how machine learning, deep learning, and reinforcement learning break through that wall; how language models turn text into economic data; and how AI agents are changing the way research is done.

AI in two roles. Artificial intelligence enters this course twice. As a **tool**, it is an extremely powerful instrument for macroeconomic research: neural networks and reinforcement learning solve models that grids cannot; language models turn text into data; and AI assistants and agents write code, search the literature and run analyses at a speed no research assistant can match. As an **object**, AI is a transformative economic force that deserves serious macroeconomic research: a new kind of capital and a new input into production, a shock to which tasks people do and what they earn, a source of market power and of new risks, and a question for fiscal policy. The same models and methods you learn here — heterogeneous agents, incomplete markets, optimal policy, and their computation — are the ones needed to study who gains and who loses from AI. The running thread on AI and inequality (Sessions 1, 9 and 10) and the economics-of-AI emphasis in the final project make this concrete.

Throughout, the course insists on one discipline: *every computed answer must be benchmarked and verified*. As AI makes generating answers cheap, verification becomes the scarce skill.

2. How the Course Is Designed: Research in the Age of AI

When an AI assistant can write most of the code, the skills that matter move up a level: knowing which question is worth asking, which model can answer it, what the equilibrium is, which method is accurate enough, which data make the model relevant, and whether the answer is right. The course is built to train exactly these skills.

- **You are the project manager.** AI executes; you decide, and you are accountable for every number. Assignments grade your decisions and the evidence behind them, not only the final answer.
- **A ladder of increasing ownership.** Homework 1 gives you the question and asks you to make the modeling, method and data decisions; Homework 2 lets you choose a route and a question of your own within a shared core; the final project is yours from question to presentation. Each homework produces a *component* that the final project reuses.
- **Several tools for one question.** Every major result is obtained at least two ways — a closed form and a grid, perturbation and projection, a classical solver and a neural network — so you learn what each method buys and when to trust it.
- **Quality control is part of the work.** You validate your own solvers against known answers, keep a validation ledger for every claim, audit a classmate's replication package, and "red-team" an AI assistant by finding and documenting its errors.
- **Learning in the open.** A course forum (anonymous or named) is where questions are asked, pitches are posted and teams form; the final mini-conference, with discussants, is where results are defended.

3. Learning Objectives

By the end of the quarter you should be able to:

1. **Theory.** Define, characterize, and interpret equilibrium in the growth model, the Bewley–Huggett–Aiyagari model, the Krusell–Smith model, overlapping-generations models, and Ramsey and Markov-perfect optimal-policy problems.
2. **Algorithms.** Design solution algorithms — value function iteration, time iteration, the endogenous grid method, perturbation, projection, nested fixed-point methods for heterogeneous-agent models — and explain their accuracy, speed, and failure modes.
3. **Implementation.** Implement these algorithms in Python and PyTorch, including neural-network and reinforcement-learning solutions of dynamic models, using AI coding assistants responsibly.
4. **Validation and research use.** Verify numerical solutions against benchmarks; use language models as measurement instruments with proper validation; and design an AI-assisted research workflow that you remain in control of.
5. **AI as an economic force.** Frame the economics of AI — AI as capital, task-based automation and augmentation, measurement of AI exposure — in quantitative macroeconomic models, and ask what it implies for inequality and policy.
6. **Research leadership.** Turn a question into a research design, choose and combine models, methods and data to answer it, check the quality of your own and others' work, and communicate the result as a paper, a talk and a reproducible package.

4. Prerequisites

The first-year Ph.D. macroeconomics sequence (or equivalent), including dynamic optimization. Programming experience is helpful but not required: Python is taught from the ground up in Session 3, and a setup lab is posted before then. Students from other fields are welcome with the instructor's consent.

5. Course Materials

- **Slides.** Short decks (about 20 minutes each; three per block) are posted on the course website before each session. The decks are self-contained.
- **Labs.** Jupyter notebooks in Python and PyTorch accompany the lectures and run on a laptop; Google Colab (or the UCR High-Performance Computing Center) can be used for GPU work.
- **Readings.** There is no required textbook. Recommended references:
  - L. Ljungqvist and T. J. Sargent, *Recursive Macroeconomic Theory*, 4th ed., MIT Press, 2018.
  - N. Stokey, R. Lucas, and E. Prescott, *Recursive Methods in Economic Dynamics*, Harvard University Press, 1989.
  - K. Judd, *Numerical Methods in Economics*, MIT Press, 1998.
  - B. Heer and A. Maussner, *Dynamic General Equilibrium Modeling*, 2nd ed., Springer, 2009.
  - Z. Feng, 《机器学习与数量宏观经济学：以PyTorch为工具》 (*Machine Learning for Quantitative Macroeconomics: Using PyTorch*), Peking University Press, 2025.
  - I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.
  - R. Sutton and A. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018.

  Session-specific readings are listed in Section 10.

6. Schedule

Each session has two blocks (A: 9:00–10:15; B: 10:35–11:50). The schedule may be adjusted as the course progresses; changes will be announced in class and on the website.

Session	Block A	Block B	Due / out
1 Sep 24	**Introduction.** Who I am and why this course; the macro research frontier; AI as a tool and as an object of macro research; how the course works.	**The optimal growth model.** From Solow to the planner; Euler equations and steady states; recursive formulation and dynamic programming; the growth model as a platform.	
2 Oct 1	**Incomplete markets.** Bewley and Huggett; Aiyagari; stationary recursive competitive equilibrium; the stationary distribution; extensions (health, entrepreneurship, portfolios).	**Aggregate risk, finite lives, policy.** Krusell–Smith; overlapping generations; Ramsey taxation, time consistency, and Markov-perfect policy.	
3 Oct 8	**Introduction to computation.** What it means to solve a model; value function iteration on a grid; Euler-equation methods and time iteration.	**Programming basics.** Python, NumPy, and PyTorch from the ground up; functions, classes, vectorization, automatic differentiation; coding with AI assistants.	HW 1 Part A out
4 Oct 15	**Numerical methods I.** Root-finding and fixed points; optimization; numerical and automatic differentiation.	**Numerical methods I (cont.).** Function approximation and interpolation; quadrature; discretizing AR(1) processes (Tauchen, Rouwenhorst); a fast and accurate stochastic growth model.	HW 1 Part B out
5 Oct 22	**Numerical methods II: perturbation.** First- and second-order perturbation; Blanchard–Kahn conditions and linear rational-expectations systems; pruning.	**Numerical methods II: projection.** Chebyshev polynomials, Smolyak grids, finite elements; collocation vs. Galerkin; the same model solved four ways, and choosing the right tool.	
6 Oct 29	**Heterogeneous agents I: the Aiyagari economy.** The household by value function iteration and by the Euler equation (time iteration, the endogenous grid method); the stationary distribution of a given policy; clearing the market by bisection; checks, and what the model misses.	**Heterogeneous agents II: aggregate risk, transitions, the life cycle.** Krusell–Smith by VFI and EGM, and Den Haan's test; an MIT-shock transition in the growth model and with a distribution; welfare along the path; the life-cycle economy; why grids and moments break down.	HW 1 due; HW 2 Part A out
7 Nov 5	**Machine learning essentials.** Learning as function approximation; neural networks; training, backpropagation, and generalization.	**Deep learning for dynamic models.** PyTorch, and a yield forecast scored against a random walk; the growth model solved on its Euler residual; shape constraints, double descent, and architecture as a constraint.	Project teams and topics

Session	Block A	Block B	Due / out
8 Nov 12	**Reinforcement learning.** RL as dynamic programming without the model; Monte Carlo, temporal-difference learning, Q-learning; policy gradients, actor–critic and deep value iteration.	**Heterogeneous agents at the frontier.** The distribution as a network input; deep equilibrium nets and DeepHAM; optimal taxation without commitment with heterogeneous agents.	HW 2 Part B out
9 Nov 19	**Language models and text as data.** From models to data; text as measurement; from words to vectors; attention and the Transformer.	**LLMs as research instruments.** Limits and validation; measuring the economic footprint of AI; retrieval-augmented generation.	Proposal due Mon Nov 16
Monday, November 23: HW 2 due. November 26: no class (Thanksgiving).			
10 Dec 3	**Agentic AI and the future of research.** How AI agents work; building a research workflow; verification, governance, and what remains human.	**Final project presentations** (mini-conference with discussants).	
Dec 10	**Final paper and replication package due** (finals week).		

7. Assessment

Component	Weight	Dates
Homework 1: from a question to a quantitative answer	15%	out Oct 8 and Oct 15; due Oct 29
Homework 2: heterogeneity and learning	15%	out Oct 29 and Nov 12; due Mon Nov 23
Final project (teams of 2–3)	40%	see below
Proposal (two pages)	5%	due Monday, Nov 16
Presentation and discussion	15%	Dec 3, Block B
Paper ($\leq$ 10 pages) and replication package	20%	due Dec 10
Participation	30%	every session

Homework and the final project: one ladder

The two homework assignments and the final project form a single sequence that teaches you to run a quantitative research project: to formulate the question yourself, choose the model, define the equilibrium, pick and validate the numerical or AI method, discipline the model with data, check the quality of your own and others' work, and present the result. Each rung gives you less scaffolding and more ownership, and each homework builds a *component* that the final project reuses. Each homework also lets you choose a *route* — a direction in which to go deeper — and includes an AI red-team item, in which you find and document an error in an AI assistant's answer.

Homework 1 (Sessions 1–5; due October 29): the representative-agent engine. You act as project manager for a common question pair — how much of U.S. business-cycle volatility TFP shocks explain, and what aggregate fluctuations cost in welfare. You write a project charter, the model and its equilibrium, validate your solver against a closed form, build your own data targets and calibration from FRED and the literature, solve the model by grid, perturbation and projection

methods, and answer both questions in a short paper, five slides and a one-command replication package. Routes: the numerical frontier, data and estimation, or fiscal policy. You finish by pitching a question of your own on the course forum.

Homework 2 (Sessions 6–9; due November 23): heterogeneity and learning. A common core — the Aiyagari economy solved as a nested fixed point, a neural-network solution validated against a classical one, a first reinforcement-learning exercise, and a first text-as-data measurement — and a route of your choice: aggregate risk and the welfare cost of cycles with heterogeneity, fiscal policy and transitions, AI solution methods, or AI and inequality with text-disciplined shocks. You also audit a classmate's Homework 1 replication package.

Each homework is submitted as a PDF, a repository that reproduces every number with one command, and an `AI_LOG.md` with a validation ledger (see Section 8). Templates are posted with each assignment. You may discuss homework with classmates, but each student writes and submits their own work and lists the people they worked with. Late homework loses 10% per day and is not accepted after solutions are posted.

Final project

In teams of two or three, you *assemble* your components to answer a question your team formulates, usually grown from the pitches. The project must combine at least one Homework 1 component and one Homework 2 component from its members — each first validated against its homework benchmark — and add what is missing: a model, a data source, or a method. It must use at least one classical and one AI/ML method cross-validated on the same object, document its AI-assisted (agentic) workflow, and include an internal audit in which one teammate re-runs another's component. Emphases include quantitative policy experiments, AI solution methods, the economics of AI, and text-based measurement.

Milestones: pitches on the forum by October 29; teams and topics by November 5; a two-page proposal by Monday, November 16 (question, components reused, what is added, division of labor, risks); presentations at the December 3 mini-conference, where each team also serves as discussant for another team; and a paper of at most ten pages with a replication package by December 10. Projects are graded on framing, integration and correctness, validation, reproducibility and provenance, quality of AI collaboration, communication, and the discussant report. The guidelines, rubric and templates are posted on the course website.

Participation

There is no examination in this course. Participation carries 30% and includes exit tickets at the end of each session, engagement in discussion — in class and on the course forum — and in the live demos, and your role as a discussant at the final-project conference.

8. Use of AI Tools

This course teaches you to use AI as a research tool, so AI assistants and agents are **encouraged** for homework and the final project. Use them the way a good project manager uses a capable but fallible team: delegate execution, keep the decisions, and check the work. Three rules apply:

1. **Disclose.** Keep an `AI_LOG.md` with each submission: the tools you used, the prompts that mattered, where the AI was wrong, and what you verified yourself.
2. **Own it.** You must be able to explain and defend every line of code and every result you submit. "The AI wrote it" is never an explanation.
3. **Verify.** Every numerical result needs a benchmark or an accuracy check, and every citation

must be checked against the original. Each homework includes an AI red-team item, and each submission includes a validation ledger linking every claim to the check that supports it.

The final project additionally documents its agentic workflow: the repository's instructions to the AI (a `CLAUDE.md` or equivalent), what was delegated, and what was verified by hand. Never commit API keys or confidential data.

9. Course Policies

Academic integrity. All work must follow the University of California, Riverside policy on academic integrity. Unauthorized collaboration, undisclosed use of AI tools, or submitting work you cannot explain is a violation of that policy and of the rules above.

Accessibility. Students who need accommodations should contact the Student Disability Resource Center and let the instructor know as early as possible so arrangements can be made.

Well-being. If personal circumstances affect your work in the course, please reach out early. UCR's Counseling and Psychological Services and other campus resources are available to support you.

Communication. Announcements and materials are posted on the course website. Questions about the material, the labs and the homework go to the course **discussion forum** on the website (sign in with your UCR e-mail), where you may post under your name or anonymously; anonymous posts are anonymous to classmates, but instructors can see who wrote them, except in the separate anonymous-feedback category. Use e-mail for private matters, and please allow up to two business days for a reply.

10. Readings by Session

Session 1 — Introduction; the growth model
Ljungqvist and Sargent (2018), chs. 3–4; Stokey, Lucas, and Prescott (1989), chs. 2–4; Acemoglu, "The Simple Macroeconomics of AI," *Economic Policy* (2025); Korinek, "Generative AI for Economic Research: Use Cases and Implications for Economists," *Journal of Economic Literature* (2023).

Session 2 — Heterogeneous agents, OLG, optimal policy
Huggett, "The Risk-Free Rate in Heterogeneous-Agent Incomplete-Insurance Economies," *JEDC* (1993); Aiyagari, "Uninsured Idiosyncratic Risk and Aggregate Saving," *QJE* (1994); Krusell and Smith, "Income and Wealth Heterogeneity in the Macroeconomy," *JPE* (1998); Heathcote, Storesletten, and Violante, "Quantitative Macroeconomics with Heterogeneous Households," *Annual Review of Economics* (2009); Diamond, "National Debt in a Neoclassical Growth Model," *AER* (1965); Chamley, *Econometrica* (1986); Judd, *Journal of Public Economics* (1985); Kydland and Prescott, "Rules Rather than Discretion," *JPE* (1977); Klein, Krusell, and Ríos-Rull, "Time-Consistent Public Policy," *REStud* (2008); Feng, "Time-Consistent Optimal Fiscal Policy over the Business Cycle," *Quantitative Economics* (2015).

Session 3 — Computation and programming
Judd (1998), ch. 12; Aruoba, Fernández-Villaverde, and Rubio-Ramírez, "Comparing Solution Methods for Dynamic Equilibrium Economies," *JEDC* (2006); QuantEcon lectures (Sargent and Stachurski), online.

Session 4 — Numerical methods I
Judd (1998), chs. 4–7; Tauchen, "Finite State Markov-Chain Approximations to Univariate and Vector Autoregressions," *Economics Letters* (1986); Kopecky and Suen, "Finite State Markov-Chain Approximations to Highly Persistent Processes," *RED* (2010); Carroll, "The Method of Endogenous Gridpoints for Solving Dynamic Stochastic Optimization Problems," *Economics Letters* (2006).

Session 5 — Numerical methods II
Blanchard and Kahn, *Econometrica* (1980); Klein, *JEDC* (2000); Schmitt-Grohé and Uribe, *JEDC* (2004); Andreasen, Fernández-Villaverde, and Rubio-Ramírez, "The Pruned State-Space System for Non-Linear DSGE Models," *REStud* (2018); Fernández-Villaverde, Rubio-Ramírez, and Schorfheide, "Solution and Estimation Methods for DSGE Models," *Handbook of Macroeconomics*, vol. 2 (2016); Judd, "Projection

Methods for Solving Aggregate Growth Models," *JET* (1992); Gaspar and Judd, "Solving Large-Scale Rational-Expectations Models," *Macroeconomic Dynamics* (1997).

Session 6 — Heterogeneous agents I and II

Aiyagari, "Uninsured Idiosyncratic Risk and Aggregate Saving," *QJE* (1994); Carroll, "The Method of Endogenous Gridpoints for Solving Dynamic Stochastic Optimization Problems," *Economics Letters* (2006); Krusell and Smith, "Income and Wealth Heterogeneity in the Macroeconomy," *JPE* (1998); Young, "Solving the Incomplete Markets Model with Aggregate Uncertainty Using the Krusell–Smith Algorithm and Non-Stochastic Simulations," *JEDC* (2010); Den Haan, "Assessing the Accuracy of the Aggregate Law of Motion in Models with Heterogeneous Agents," *JEDC* (2010); Reiter, *JEDC* (2009); Auclert, Bardóczy, Rognlie, and Straub, "Using the Sequence-Space Jacobian to Solve and Estimate Heterogeneous-Agent Models," *Econometrica* (2021); Conesa, Kitao, and Krueger, "Taxing Capital? Not a Bad Idea After All!," *AER* (2009). *On Krusell–Smith and transitions:* Maliar, Maliar, and Valli, "Solving the Incomplete Markets Model with Aggregate Uncertainty Using the Krusell–Smith Algorithm," *JEDC* (2010); Auerbach and Kotlikoff, *Dynamic Fiscal Policy* (1987).

Session 7 — Machine learning and deep learning for dynamic models

Goodfellow, Bengio, and Courville (2016), chs. 6–8; Maliar, Maliar, and Winant, "Deep Learning for Solving Dynamic Economic Models," *Journal of Monetary Economics* (2021); Fernández-Villaverde, Hurtado, and Nuño, "Financial Frictions and the Wealth Distribution," *Econometrica* (2023).

Session 8 — Reinforcement learning and heterogeneous agents

Sutton and Barto (2018), chs. 3–6 and 13; Azinovic, Gaegauf, and Scheidegger, "Deep Equilibrium Nets," *International Economic Review* (2022); Han, Yang, and E, "DeepHAM: A Global Solution Method for Heterogeneous Agent Models with Aggregate Shocks," arXiv:2112.14377.

Session 9 — Language models and measuring AI

Gentzkow, Kelly, and Taddy, "Text as Data," *Journal of Economic Literature* (2019); Baker, Bloom, and Davis, "Measuring Economic Policy Uncertainty," *QJE* (2016); Vaswani et al., "Attention Is All You Need," *NeurIPS* (2017); Eloundou, Manning, Mollick, and Rock, "GPTs Are GPTs," *Science* (2024); Acemoglu and Restrepo, "The Race between Man and Machine," *AER* (2018).

Session 10 — Agentic AI

Korinek (2023), revisited; Handa et al., "Which Economic Tasks Are Performed with AI? Evidence from Millions of Claude Conversations," arXiv:2503.04761 (2025).